

MEC-109
Research Methods
in Economics

Block

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RESEARCH DESIGN AND MEASUREMENT

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Print Production

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SOSS, IGNOU, New Delhi

October, 2015

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ISBN-978-81-266-

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Further information on Indira Gandhi National Open University courses may be obtained from the University's office at Maidan Garhi. New Delhi-110 068.

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi by the Director, School of Social Sciences.

Laser Typeset by : Tessa Media & Computers, C-206, A.F.E.-II, Okhla, New Delhi

Printed at :

BLOCK 2 RESEARCH DESIGN AND MEASUREMENT

Research design is a logical structure of an enquiry and its formulation is guided and determined by the research questions raised in the problem under investigation. Apart from specifying the logical structure of data, research design also test and eliminate alternative explanation. Broadly, the observational design sampling design and statistical design are covered in Research Design. The various attributes of people, objects or concepts are being increasingly included in explanation of human behaviour in Economics. Hence, these individual traits, attitudes need to measure for deeper analysis. Research Design and measurement issues, therefore constitute the theme of this block. The block comprises of 3 Units.

Throwing light on the concept of research design and various types of research, **Unit 6** covers the rationale and forms of mixed methods research with illustration of three case studies.

Unit 7 provides comprehensive knowledge of all the elements for collecting the quantitative data for research study. It covers the methods and tools of data collection and the various issues related to sampling design.

Unit 8 deals with the various measurement scales, scaling techniques, and criteria for good measurement and sources of errors in measurement.

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UNIT 6 RESEARCH DESIGN AND MIXED METHODS RESEARCH

Structure

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- 6.1 Introduction
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- 6.11 Answers or Hints to Check Your Progress Exercises

6.0 OBJECTIVES

After going through this Unit, you will be able to:

- state the various types of research;
- explain the concept and meaning of research design;
- make a distinction between research design and research methods;
- delineate the characteristics of quantitative research and qualitative research;
- discuss the purposes of mixed methods research; and
- appreciate the application of mixed methods research in conducting research studies in social sciences in general and economics in particular.

6.1 INTRODUCTION

Due to confusion or lack of clarity between research design and research methods, we often use these two terms interchangeably. As a result, the research design is evaluated on the basis of strengths and weakness of the method rather its ability to draw relatively unambiguous conclusions. Hence

efforts have been made here to throw light on the distinction between these two terms. Traditionally two approaches – quantitative research (QN) associated with post positivist paradigm and qualitative research (QL) associated with interpretative paradigm have been followed in social sciences. However, in recent years increasing reliance on mixed methods research by combining quantitative and qualitative methods in a single research project has been observed. Hence in this unit, the concept of mixed method research, its types and application have been covered. Application of mixed method research in a subject like economics which is essentially quantitative in nature have been illustrated by case studies. Before taking up the issue of research design and research method, it is desirable to have an idea about various types of research. Hence let us begin to introduce the various types of research.

6.2 TYPES OF RESEARCH

There are different ‘types of research’ depending on their nature and field of specialization. While a classification based on a dichotomous distinction (like, theoretical/applied, descriptive/analytical, conceptual/empirical, etc.) is possible, it is necessary to recognize that there may be overlapping features rendering such a classification less perfect to that extent. It is, therefore, useful to first of all take a look at the underlying basic features so as to be able to identify the different types of research within their respective connotations and usage.

A variable which can be measured can take values or scores either in quantitative or qualitative terms. For instance, yield of an agricultural output, height/weight of individuals, etc. can be measured and expressed in quantitative terms. On the other hand, characteristics like one’s feelings or opinion (e.g., good/bad, male/female, agreeing/disagreeing, yes/no, etc.) are attributes on which a choice can be expressed (and a score assigned) depending on the possible alternatives or choices. Thus, variables like the performance of an artist, respondent’s gender, etc. on which responses can only be categorized or grouped are ‘**qualitative variables**’. From this angle, variables on which data can be collected and expressed in quantified terms may be called as ‘**quantitative variables**’. Even variables which are expressed in qualitative terms can be counted whereupon their total numbers become quantified expressions. A popular example of developmental economics which ranks different countries based on their status attained in a number of areas/variables (some of which like the political freedom/choice enjoyed by the country, though expressed in some quantitative measure, is qualitative in its nature) is the human development index. In actual research (particularly research in the field of social sciences), it is important to note that variables of both the nature normally co-exist. In view of this, the analysis pertaining to each type of variable/data needs different methods and techniques.

The type of research would also vary depending on the objectives of the study. Research design varies with the type of research one likes to pursue. With this background, we can now proceed to know the distinctions between different types of research.

6.2.1 Theoretical and Applied Research

Research can either be theoretical or applied. The former i.e., theoretical research can also be considered as ‘fundamental research’ as its outcome serves

as a foundation for all subsequent development in the field. Fundamental (or basic) research mainly concerns with formulation of theory with knowledge perceived as an end in itself. It, thus, aims at obtaining knowledge of the logical processes involved in a phenomenon. It pertains to the quest for knowledge about a phenomenon without concern for its practical use. Such a research may either verify the old theory or establish a new one. For example, fundamental research in economics may consist of research to develop and improve economic theories or evolve quantitative techniques to measure parameters such as multiplier effect, elasticity of demand and supply, etc. Fundamental research is essentially positive in nature.

Applied research, on the other hand, aims at finding a solution for a problem facing society or industry. It is, thus, applied to practical situations or contexts. While pure research discovers principles and laws, applied research discovers ways of applying them to solve specific problems. It is useful to test the theories developed empirically and can as a result also contribute to improving the tools and techniques of measurement. Basic research, therefore, can be treated as building blocks for applied research. Illustrations of applied research in economics can be measurement of poverty, employment, rural development, agriculture, environment, etc.

6.2.2 Descriptive and Explanatory Research

Descriptive research describes a situation, events or social systems. It aims to describe the state of affairs as it exists. Surveys and fact-finding enquiries of different kinds are part of descriptive research. Survey methods of all kinds including comparative and correlational methods are used in descriptive research studies. A survey of socio-economic conditions of rural/urban labour is an area of descriptive research. In descriptive research studies, the researchers have no control over variable. They can report only what has happened or is happening. Descriptive research deals with questions like ‘how does X vary with Y?’ or ‘how does malnutrition vary with age and sex?’, etc.

Explanatory research, aims at establishing the cause and effect relationship. The researcher uses the facts or information already available to analyse and make a critical evaluation of the data/information. An example of explanatory research is: ‘whether increase in agricultural productivity is explained by improved rural roads?’

6.2.3 Quantitative and Qualitative Research

Quantitative Research is the systematic and scientific investigation of quantitative properties and phenomena and their relationships. The objective of quantitative research is to develop mathematical models, and to test hypotheses. Based on ontological and epistemological premises, quantitative research is characterized by the following attributes:

- A belief in a single reality.
- The pursuit of identifying universal laws.
- Separation of knower (researcher) from known (researched).
- The possibility and necessity of value free research.
- Pursuit of universal laws (findings) beyond the limit of research/social context.

- The tendency to work with large, representative sample.
- An emphasis on deductive research via falsifiable hypothesis.
- Formal hypothesis testing.

In contrast, **Qualitative Research** captures a set of purposes associated with meaning and interpretation. Stress is laid on the socially constructed nature of reality, the intimate relationship between the research and researched and situational constraints that shape the enquiry. The key attributes associated with qualitative research are:

- A belief in a constructed reality, multiple realities, co-existence realities.
- An interdependence between the knower and the known.
- Impossibility to separate the researcher from the research subject.
- Heavy role of context in research process.
- The impossibility to generalize research findings beyond the limits of the immediate context.
- Non-separation of cause and effect.
- The explicit focus on inductive, exploratory research approaches.
- The tendency to work with small and purposely chosen sampling.
- Analyses holistic system.

6.2.4 Conceptual and Empirical Research

Conceptual research is related to abstract ideas or theory. It is generally used by philosophers and thinkers to develop new concepts or to reinterpret the existing theories.

Empirical research relies on experience or observation. It is data based. It is subject to verification by observation and experiment. This type of research is particularly useful when validation or verification of an aspect is required.

6.2.5 Other Types of Research

Research may be exploratory or formalized. **Exploratory** research aims at developing the hypothesis rather than testing a pre-conceived hypothetical contention or notion. **Formalised** research studies deal with a definitive structure within which specific hypotheses are tested. **Historical research** utilizes existing documents to study events of the past. Research can also be experimental or evaluative. **Experimental research** aims at identifying the causal factors by means of experiments. In **evaluative research**, the cost effectiveness of a programme is examined. Such research addresses the question of the efficiency of a programme and are useful in taking policy decisions on issues like whether the programme is effective and/or needs modification or re-orientation? Whether it should be continued?

Action research is another type of research. It deals with real world problems aimed at finding out practical solutions or answers to them. It gathers feedback which is then used to generate ideas for improvement. You will find details on Action Research in Unit 20.

Check Your Progress 1

- 1) Distinguish between quantitative variable and qualitative variable.

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- 2) State the main objectives of applied research?

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- 3) In what sense quantitative research is different from qualitative research?

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- 4) What is the distinction between explanatory and descriptive research?

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6.3 RESEARCH DESIGN

Research design is a logical structure of an enquiry. Given the research question or theory, what type of evidence is needed to answer the question (or to test the theory) in a convincing way – constitutes the essence of the research design. Let us use an analogy to understand the term ‘research design’. While constructing a building, the first decision to be arrived at is: whether we need a high rise office building, a factory, a school or a residential apartment etc.? Until this is decided, we cannot sketch a plan and order material or setting critical dates for completion of the project dates. Similarly, a social researcher needs to be clear about the research questions and then the research design will flow from the research questions. The function of a research design is to ensure that the evidence obtained enables us to answer the initial research questions as unambiguously as possible. Obtaining relevant evidence entails specifying the type of evidence we need to answer the research question, to test a theory, to evaluate a programme or to accurately describe some phenomenon. The issues of sampling, method of data collection (e.g. questionnaire, observation, document analysis), design of questionnaire etc. are all subsidiary to what constitute the evidences that need to collect to answer the research questions.

Thus the research design ‘deals with a logical problem and not a logistical problem. Apart from specifying the logical structure of the data, it also aims to test and eliminate alternative explanation of results.

The research design comprises of the following components:

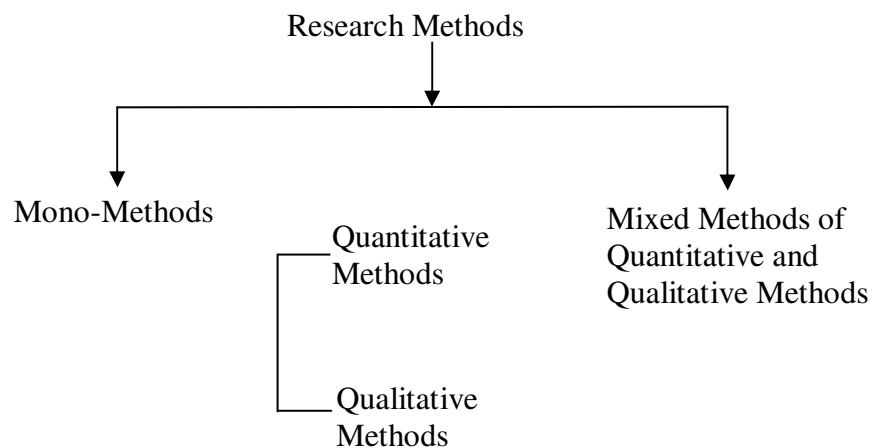
- i) the sampling design; (the type of sampling method i.e. random or non random sampling);
- ii) the statistical design (i.e. the sample size and the method to draw the sampling to be adopted);
- iii) the observational design (i.e. the instrument of collection of data);
- iv) the operational design i.e. the specific details by which the procedure in (i), (ii) & (iii) above are to be carried out.

6.4 RESEARCH DESIGN VS. RESEARCH METHODS

Research design is different from the research methods. The research methods are made of data collection and techniques of data analysis whereas the research design is a logical structure of an enquiry. There is nothing intrinsic about any research design that requires a particular method of data collection. How the data are collected is irrelevant to the logic of the research design. Confusing research designs with research methods leads to poor evaluation of designs. For example equating cross sectional designs with questionnaires or case studies with participant observation result in evaluation of research designs against the strengths and weakness of the method rather than their ability to draw relatively unambiguous conclusion or to select between rival plausible hypothesis.

6.5 RESEARCH METHODS

Research methods as explained in Unit 1 refers to tools and techniques of data collection and data analysis. Broadly Research Methods are put under two categories: (i) Mono Methods, (ii) Mixed Methods. Within mono methods, there are two approaches (i) quantitative (QN) methods (ii) qualitative (QL) methods. Since quantitative methods are associated with positivist and post-positivist paradigm, quantitative method is also termed as (QN) method research. Similarly by virtue of the fact that QL methods are associated with interpretativist paradigm and critical theory paradigm, research conducted by the qualitative method is also termed as qualitative methods research.



6.5.1 Quantitative Methods

The researcher using this approach tests theory through empirical observations and inclines to establish cause and effect relationship. Well predesigned structured questionnaire and structured interviews are resorted as tool to collect the data. The variables that can be measured are emphasized. The descriptive, analytical and inferential statistical techniques are used to analyse the data. Experimental research design is used to make the logical structure of the research study. Generalization of laws and their replication is emphasized.

6.5.2 Qualitative Methods

Under qualitative approach, a researcher starts with observation as a basis for generating theory and concentrates on meaning of observations. He studies events as they occur in naturalistic settings. He allows interview topics to emerge during conversation and listens to others' interpretation and perspectives.

Under qualitative method, open-end questionnaire, key informants, group discussion and unstructured interviews, documents, interview transcripts etc. are used as tools to collect data. Content analysis, Case study, Action research participatory method, Cluster analysis, Factor analysis, Correspondence analysis, Context analysis, are used in analyzing qualitative data. You will find details of these techniques in Unit 13, 14, 15, 16 and 17 of Block 4 and 18, 19 and 20 of Block 5 of this course.

6.5.3 Mixed Methods

The combination of at least one qualitative and at least one quantitative component related to measurement scale, tools of data collection or data analysis technique in a single research study/project is known as mixed methods research. For example it can be in the following forms:

- Employment of more than one measurement procedure in different kind of situations such as different ways of measuring levels of job satisfaction.
- Employment of two or more methods of data collection. For example we may employ observation, interview and questionnaire to collect the data about decent work.
- Employment of two or more data analysis techniques for example, content analysis and factor analysis.

Mixed methods research has gained a tremendous popularity in social, behavioural and related sciences in recent years. The rationale for mixed method design research is to take the best of qualitative (QL) and quantitative methods (QN) and combine them. However, many debates on mixed method research design are based on methodological arguments due to the ways QN method is linked to positivist paradigm and QL method to Interpretative and critical theory paradigm.

As discussed above, keeping in view the diametrical oppositional attributes of QL and QN methods, one may argue that these methods are not compatible to each other and in such a situation how a truce between two kinds of approaches be useful to conduct the meaningful research?

Notwithstanding the philosophical debate between QN and QL approach. Mixed methods research designs are justified primarily by exploiting the